

# From Direct Algebraic Generating Functions to Elliptic Sigma and Theta Representations of Somos–4 Sequences

A Constructive Workflow

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## Abstract

A direct five-parameter algebraic generating function can be constructed whose Hankel transform follows a prescribed nondegenerate Somos–4 orbit. Its discriminant contains the genus-one curve

$$E_J : \quad y^2 = (X^2 - JX + \tau)^2 - 4X.$$

The elliptic-function theory behind Somos–4 sequences is classical, but the practical passage from concrete recurrence data to a numerical sigma- or theta-function representation is distributed across several sources and conventions. This paper assembles those ingredients into one constructive and reproducible workflow:

$$(\lambda, m, r, \eta, \tau) \mapsto J \mapsto E_J \mapsto (g_2, g_3) \mapsto (\omega_1, \omega_2) \mapsto (z_0, \delta) \mapsto \sigma \mapsto \theta_1.$$

The emphasis is on normalization choices, computational checks, and an executable PARI/GP implementation. No new proof of the classical sigma-function representation is intended; the contribution is the unification of the scattered constructions into an applicable end-to-end procedure beginning with the algebraic generating function.

## 1 Introduction

The relation between Somos–4 recurrences, elliptic curves, and elliptic functions is well established through the work of Gale, Somos, van der Poorten, Swart, Hone, and others. In particular, generic Somos–4 orbits admit representations in terms of the Weierstrass sigma function. Nevertheless, a reader beginning with a concrete sequence still faces a substantial practical problem: the relevant constructions occur in different normalizations, coordinate systems, and parametrizations, and the intermediate steps needed for a numerical implementation are rarely presented together. The purpose of this paper is not to prove a new elliptic-function theorem. Rather, it is to make the established theory directly usable. Our starting point is the direct algebraic generating function constructed in [1]. Its discriminant naturally produces a genus-one curve associated with the Somos–4 data. Beginning with this curve, we describe how to obtain a Weierstrass model, compute its period lattice, identify the point corresponding to one recurrence step, determine the Abel–Jacobi parameters, and finally write explicit sigma- and theta-function representations. The paper is therefore organized as a mathematical workflow rather than as a theorem–proof sequence. Classical facts are stated with references and are not reproved. Special attention is paid to the information required at each stage, the conventions that must remain consistent, and the checks that detect an incorrect branch, period, sign, or scaling. PARI/GP fragments accompany the construction, and a complete script is planned for the appendix.

## 1.1 Scope and contribution

This paper provides:

- a single route from the five input parameters to an elliptic curve;
- an explicit inventory of the normalization choices needed to pass from a quartic model to sigma and theta functions;
- computational checkpoints at every stage;
- worked examples comparing recurrence, algebraic-generating-function, sigma-function, and theta-function evaluations; and
- an end-to-end PARI/GP workflow suitable for adaptation.

It does not claim a new proof of the classical Somos–4 sigma formula, a new integrability result, or a new theorem on elliptic curves.

## 2 Input data and the direct algebraic generating function

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be the algebraic generating function constructed in [1]. It satisfies

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0,$$

where the polynomials  $A, B, C$  depend explicitly on

$$(\lambda, m, r, \eta, \tau).$$

The associated Hankel transform is initialized by

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta,$$

and follows the corresponding Somos–4 orbit with recurrence parameter  $\tau$  in the normalization of [1]. The discriminant factors as

$$B(z)^2 - 4A(z)C(z) = (\lambda^2 \eta m r^2 z)^2 \left( (1 - Jz + \tau z^2)^2 - 4z^3 \right),$$

where

$$J = \frac{\lambda \eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m \eta} + \frac{\tau m r}{\lambda \eta}. \quad (1)$$

Removing the square prefactor and applying the reciprocal change of variable suggested by the residual polynomial gives the quartic model

$$E_J : \quad y^2 = (X^2 - JX + \tau)^2 - 4X. \quad (2)$$

The subsequent workflow begins with equations (1) and (2).

### 3 What information is already contained in the discriminant?

The factorization above provides the conceptual bridge between the algebraic and transcendental descriptions. Once the irrelevant square factor is removed, the generating-function discriminant supplies a quartic genus-one model depending only on  $(J, \tau)$ . Thus the five input parameters determine the initial placement and normalization of the orbit, while the curve itself is encoded by the reduced pair  $(J, \tau)$ . This observation should be interpreted carefully. The discriminant already contains the elliptic curve, but it does not by itself fix every choice needed for a numerical theta representation. One must still:

1. choose and verify a birational map to a Weierstrass model;
2. adopt an ordering and orientation for the periods;
3. identify the point implementing one Somos step;
4. choose Abel–Jacobi lifts of the initial point and translation;
5. determine the elementary exponential factors; and
6. check the result against the original recurrence.

In this precise sense, the theta function is hidden in the discriminant, but extracting it requires a controlled sequence of normalizations.

## 4 The constructive workflow

### 4.1 Workflow Step 1: Compute the invariant and construct the quartic

Given  $(\lambda, m, r, \eta, \tau)$ , compute  $J$  using equation (1) and form the quartic equation (2). Before proceeding, test that the quartic is nonsingular. Equivalently, its polynomial discriminant must be nonzero. Degenerate inputs require a separate rational or singular analysis and are outside the principal workflow.

**Checkpoint.** Expand the two sides of the generating-function discriminant identity and verify exact equality in the coefficient field.

### 4.2 Workflow Step 2: Convert the quartic to Weierstrass form

Choose as origin the point at infinity characterized by  $y/X^2 \rightarrow 1$ . Put

$$U = \frac{y + X^2 - JX + \tau}{2}, \quad V = U(2X - J) - 1. \quad (3)$$

Since the quartic equation implies

$$U(U - (X^2 - JX + \tau)) = -X,$$

a direct elimination of  $X$  and  $y$  yields

$$V^2 = 4U^3 + (J^2 - 4\tau)U^2 + 2JU + 1. \quad (4)$$

The shift

$$u = U + \frac{J^2 - 4\tau}{12}, \quad v = V \quad (5)$$

therefore gives the Weierstrass model

$$v^2 = 4u^3 - g_2u - g_3, \quad (6)$$

where, with  $a = J^2 - 4\tau$ ,

$$g_2 = \frac{a^2}{12} - 2J, \quad g_3 = -\frac{a^3}{216} + \frac{aJ}{6} - 1. \quad (7)$$

The inverse affine formulas are equally simple. First set

$$U = u - \frac{J^2 - 4\tau}{12}, \quad V = v. \quad (8)$$

For  $U \neq 0$ ,

$$X = \frac{V + JU + 1}{2U}, \quad y = 2U - X^2 + JX - \tau. \quad (9)$$

Substitution verifies both compositions identically on their affine charts. The denominator  $U$  vanishes at the two cubic points  $(U, V) = (0, 1)$  and  $(0, -1)$ . In the projective correspondence these are two points omitted by the affine inverse chart. The chosen origin  $y/X^2 \rightarrow 1$  maps to the cubic point at infinity. The other quartic point at infinity maps to  $(U, V) = (0, 1)$ , while  $(U, V) = (0, -1)$  maps to the finite quartic point  $(X, y) = (0, -\tau)$ . Thus the sole affine exceptional set of the inverse formula is  $U = 0$ ; it is resolved by using the projective models.

The polynomial discriminant of the quartic in  $X$  is

$$\text{disc}_X((X^2 - JX + \tau)^2 - 4X) = 256\Delta, \quad (10)$$

where

$$\Delta = g_2^3 - 27g_3^2 = J^4\tau + J^3 - 8J^2\tau^2 - 36J\tau + 16\tau^3 - 27. \quad (11)$$

Consequently, over a field of characteristic zero, the birational elliptic model is nonsingular precisely when  $\Delta \neq 0$ ; no additional parameter values are excluded.

**Checkpoint.** Reduce the identity

$$V^2 - (4U^3 + (J^2 - 4\tau)U^2 + 2JU + 1)$$

modulo  $y^2 - (X^2 - JX + \tau)^2 + 4X$  and verify that the remainder is zero. Verify the two inverse compositions, the discriminant identity equation (10), and

$$j = 1728 \frac{g_2^3}{\Delta}.$$

### 4.3 Workflow Step 3: Compute and normalize the period lattice

Let  $e_1, e_2, e_3$  be the distinct roots of

$$4u^3 - g_2u - g_3 = 0.$$

They are the branch values of the double covering

$$v^2 = 4(u - e_1)(u - e_2)(u - e_3).$$

A period lattice is obtained by integrating the holomorphic differential  $du/v$  over an oriented homology basis:

$$2\omega_1 = \oint_{\gamma_1} \frac{du}{v}, \quad 2\omega_2 = \oint_{\gamma_2} \frac{du}{v}.$$

The cycles are oriented and, if necessary, replaced by an equivalent basis so that

$$\operatorname{Im}\left(\frac{\omega_2}{\omega_1}\right) > 0.$$

We then set

$$\tau_{\text{mod}} = \frac{\omega_2}{\omega_1}.$$

Only the lattice is intrinsic; the ordered pair  $(2\omega_1, 2\omega_2)$  depends on the chosen homology basis. This choice must therefore be retained unchanged in the later definitions of elliptic logarithms, sigma functions, and theta functions.

For real input data with  $\Delta > 0$ , the three roots are real and may be ordered as

$$e_1 > e_2 > e_3.$$

In this case define

$$k^2 = \frac{e_2 - e_3}{e_1 - e_3}, \quad k'^2 = 1 - k^2 = \frac{e_1 - e_2}{e_1 - e_3}.$$

With the positive real square root of  $e_1 - e_3$ , a convenient rectangular half-period basis is

$$\omega_1 = \frac{K(k)}{\sqrt{e_1 - e_3}}, \quad \omega_2 = \frac{iK(k')}{\sqrt{e_1 - e_3}}, \quad (12)$$

where

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}$$

is the complete elliptic integral of the first kind. Consequently,

$$\tau_{\text{mod}} = i \frac{K(k')}{K(k)}$$

lies in the upper half-plane.

For general complex parameters, or for real parameters with only one real root, it is preferable to compute the periods from integrals over a chosen pair of branch cuts rather than to impose the ordering used in equation (12). The resulting basis is then normalized by an element of  $\operatorname{SL}_2(\mathbb{Z})$  so that  $\operatorname{Im}(\tau_{\text{mod}}) > 0$ . A direct elliptic-curve period routine provides an independent numerical check, but the complete-integral and cycle-integral descriptions are taken as the primary mathematical definition.

**Checkpoint.** First verify numerically that

$$4(u - e_1)(u - e_2)(u - e_3) = 4u^3 - g_2u - g_3.$$

Then confirm

$$\operatorname{Im}(\tau_{\text{mod}}) > 0.$$

Finally reconstruct the invariants from the normalized lattice and check that they agree with the original  $g_2$  and  $g_3$  to the working precision. In the three-real-root case, also compare the periods from equation (12) with periods obtained by direct numerical integration of  $du/v$ .

#### 4.4 Workflow Step 4: Determine the elliptic translation point

Introduce the gauge-invariant recurrence variables

$$x_n = \frac{H_{n-1}H_{n+1}}{H_n^2}. \quad (13)$$

The Somos-4 recurrence implies the second-order QRT map

$$x_{n+2}x_nx_{n+1}^2 = x_{n+1} + \tau, \quad x_{n+2} = \frac{x_{n+1} + \tau}{x_nx_{n+1}^2}, \quad (14)$$

and the quantity

$$x_nx_{n+1} + \frac{1}{x_n} + \frac{1}{x_{n+1}} + \frac{\tau}{x_nx_{n+1}} \quad (15)$$

is independent of  $n$ . Its value is precisely  $J$  from equation (1). Indeed, for the initial data of section 2,

$$x_2 = \frac{\lambda r}{m^2}, \quad x_3 = \frac{m\eta}{r^2},$$

and substituting these values into equation (15) gives equation (1).

The invariant biquadratic becomes the quartic equation (2) under

$$X_n = x_nx_{n+1}, \quad y_n = x_n - x_{n+1}. \quad (16)$$

The sign convention in equation (16) is essential: replacing  $y_n$  by  $-y_n$  reverses the direction of the translation point below. Applying equations (3) and (5) gives

$$\begin{aligned} U_n &= \frac{y_n + X_n^2 - JX_n + \tau}{2}, & V_n &= U_n(2X_n - J) - 1, \\ u_n &= U_n + \frac{J^2 - 4\tau}{12}, & Q_n &= (u_n, V_n) \in E_J. \end{aligned} \quad (17)$$

With the group law on equation (6) and the point at infinity as identity, one QRT step is

$$Q_{n+1} = Q_n + P, \quad P = \left( \frac{J^2 - 4\tau}{12}, 1 \right). \quad (18)$$

Thus the translation point is determined directly by  $(J, \tau)$  and does not have to be recovered numerically from several iterates. A direct check uses the next QRT value from equation (14), constructs  $Q_{n+1}$  from equations (16) and (17), and verifies  $Q_{n+1} - Q_n = P$  by the Weierstrass group law. The formula is valid on the open chart where the quantities displayed above are defined; exceptional or vanishing recurrence values are handled by the projective curve and continuation in the parameters.

There is also a continued-fraction interpretation. On the quartic model, the two points at infinity differ by a fixed divisor class. Advancing one line of the genus-one continued fraction translates the Jacobian by that class. With the origin chosen in Workflow Step 2, the second point at infinity maps to  $(U, V) = (0, 1)$ , hence to the point  $P$  in equation (18). The QRT and continued-fraction routes therefore give the same increment, rather than two independently normalized candidates.

For the classical data  $H_1 = H_2 = H_3 = H_4 = 1$  and  $\tau = 1$ , one has  $J = 4$  and

$$E_J: \quad v^2 = 4u^3 - 4u + 1, \quad P = (1, 1).$$

Furthermore,  $x_2 = x_3 = 1$  and equation (14) gives  $x_4 = 2$ . The corresponding points are

$$Q_2 = (0, 1), \quad Q_3 = (-1, -1),$$

and the group law gives  $Q_3 - Q_2 = (1, 1) = P$ . On the quartic side,  $P$  is the image of the second point at infinity, so the continued-fraction divisor shift yields the same point.

**Checkpoint.** For several consecutive pairs  $(x_n, x_{n+1})$ , verify equation (15), map the states to  $Q_n$ , and check  $Q_{n+1} - Q_n = P$ . As a quick structural test, confirm directly that  $P$  lies on equation (6); in the intermediate coordinates it is the point  $(U, V) = (0, 1)$ .

#### 4.5 Workflow Step 5: Compute Abel–Jacobi coordinates

Map  $Q_0$  and  $P$  to the Weierstrass model. Choose lifts  $z_0$  and  $\delta$  in  $\mathbb{C}$  satisfying

$$(\wp(z_0), \wp'(z_0)) = Q_0, \quad (\wp(\delta), \wp'(\delta)) = P,$$

or, equivalently,

$$z_0 = \int_{\infty}^{Q_0} \frac{du}{v}, \quad \delta = \int_{\infty}^P \frac{du}{v} \pmod{\Lambda}.$$

The lifts are not unique. Their representatives should be reduced to a stated fundamental parallelogram, with signs fixed by the  $v$ -coordinates.

**Checkpoint.** For several integers  $n$ , verify that

$$(\wp(z_0 + n\delta), \wp'(z_0 + n\delta))$$

agrees with the Weierstrass image of  $Q_n$ .

#### 4.6 Workflow Step 6: Fix the sigma-function normalization

In the convention used here, the classical sigma-function solution of a nondegenerate Somos–4 recurrence is

$$H_n = AB^n \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}. \quad (19)$$

The sign of  $\delta$  is fixed in Workflow Steps 4 and 5 by requiring that  $Q_{n+1} = Q_n + P$  and  $(\wp(\delta), \wp'(\delta)) = P$ . The indexing is fixed by

$$(\wp(z_0 + n\delta), \wp'(z_0 + n\delta)) = Q_n. \quad (20)$$

In particular, the initial recurrence data determine  $Q_2$  from  $x_2 = \lambda r/m^2$  and  $x_3 = m\eta/r^2$ . If  $z_2$  is the chosen elliptic logarithm of  $Q_2$ , then the lift appearing in equation (19) is

$$z_0 = z_2 - 2\delta \pmod{\Lambda}. \quad (21)$$

This shift is essential: using  $z_2$  itself as  $z_0$  would displace the sigma formula by two recurrence indices.

For brevity, put

$$S_n = \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}. \quad (22)$$

Assuming the displayed sigma factors are nonzero, the first two initial conditions  $H_1 = \lambda$  and  $H_2 = m$  determine the elementary gauge constants uniquely:

$$B = \frac{mS_1}{\lambda S_2}, \quad A = \frac{\lambda^2 S_2}{mS_1^2}. \quad (23)$$

Equivalently,

$$B = \frac{m \sigma(z_0 + \delta) \sigma(\delta)^3}{\lambda \sigma(z_0 + 2\delta)}, \quad (24)$$

$$A = \frac{\lambda^2 \sigma(z_0 + 2\delta)}{m \sigma(\delta)^2 \sigma(z_0 + \delta)^2}. \quad (25)$$

The remaining initial values do not supply two further independent gauge constants. Instead, they are consistency conditions for the chosen curve, translation point, elliptic logarithms, and indexing:

$$r = AB^3S_3, \quad \eta = AB^4S_4. \quad (26)$$

Thus all four initial values are used: the first two fix  $A$  and  $B$ , while the third and fourth test the geometric data and the branch choices.

The exponent  $n^2$  in equation (19) is also fixed by the sigma addition formula

$$\frac{\sigma(z+\delta)\sigma(z-\delta)}{\sigma(z)^2\sigma(\delta)^2} = \wp(\delta) - \wp(z). \quad (27)$$

Indeed, substitution of equation (19) into  $x_n = H_{n-1}H_{n+1}/H_n^2$  cancels  $A$  and  $B$  and gives

$$x_n = \frac{\sigma(z_0 + (n-1)\delta)\sigma(z_0 + (n+1)\delta)}{\sigma(z_0 + n\delta)^2\sigma(\delta)^2} = \wp(\delta) - \wp(z_0 + n\delta). \quad (28)$$

This identity links the sigma normalization directly to the QRT variables and provides a sensitive check of the sign and index conventions.

**Checkpoint.** First verify equation (26) at the working precision. Then compare values from equation (19) with exact recurrence values for a range of positive and, where defined, negative indices. Independently check equation (28) against the rational QRT iterates; this detects an incorrect sign of  $\delta$ , an index shift in  $z_0$ , or an inconsistent period representative.

#### 4.7 Workflow Step 7: Convert the sigma formula to theta form

For the chosen period basis, use

$$\sigma(z) = \exp\left(\frac{\eta_1 z^2}{2\omega_1}\right) \frac{2\omega_1}{\pi\theta_1'(0 \mid \tau_{\text{mod}})} \theta_1\left(\frac{\pi z}{2\omega_1} \mid \tau_{\text{mod}}\right), \quad (29)$$

with  $\eta_1 = \zeta(\omega_1)$ . Substitution into equation (19) collects the exponential factors into constants of the form

$$K_0 q_0^{n^2} r_0^n,$$

leading to

$$H_n = K_0 q_0^{n^2} r_0^n \theta_1(A_0 n + B_0 \mid \tau_{\text{mod}}). \quad (30)$$

All constants in equation (30) must be given in terms of  $\omega_1, \eta_1, z_0, \delta, A, B$  and the theta normalization used by the software.

**Checkpoint.** Evaluate equation (19) and equation (30) independently and compare them before comparing either expression with the recurrence.

## 5 Algorithm summary

**Step 1.** Read  $(\lambda, m, r, \eta, \tau)$  and test the required nonvanishing assumptions.

**Step 2.** Compute  $J$  from equation (1).

**Step 3.** Construct  $E_J$  and test nonsingularity.

**Step 4.** Compute a birational Weierstrass model and retain both maps.

- Step 5.** Compute  $(g_2, g_3)$  and check curve invariants.
- Step 6.** Compute  $(\omega_1, \omega_2)$  and normalize the period basis.
- Step 7.** Construct consecutive QRT states and recover the fixed point  $P$ .
- Step 8.** Compute and normalize  $z_0$  and  $\delta$ .
- Step 9.** Determine the constants in the sigma representation from the initial data.
- Step 10.** Convert the result to theta form using equation (29).
- Step 11.** Validate all four descriptions: recurrence, generating function, sigma form, and theta form.

## 6 Numerical examples

### 6.1 The classical Somos–4 sequence

For the classical seed

$$(\lambda, m, r, \eta, \tau) = (1, 1, 1, 1, 1),$$

the companion script computes the invariant

$$J = \text{somos}_J(1, 1, 1, 1, 1) = 4, \quad (g_2, g_3, \Delta) = (4, -1, 37).$$

The corresponding quartic model is therefore

$$y^2 = (X^2 - 4X + 1)^2 - 4X,$$

and the period lattice computed by PARI/GP at 80 decimal digits of working precision is

$$\omega_1 = 2.9934586462319596298, \quad \omega_2 = -2.4513893819867900609i.$$

Under the normalization used by `ellperiods`, the basis is chosen with the standard positive-imaginary convention for the lattice ratio, and the Abel–Jacobi parameters returned by the script are

$$z_0 = -2.7887781458561870232 + 1.2256946909933950304i, \quad \delta = 1.8591854305707913488.$$

The sigma-normalization constants produced by the same run are

$$A = -0.23674493915977148 + 0.06797467161582982i, \quad B = 1.24599348339577469 + 3.55401276446268772i.$$

With these values the direct sigma evaluation gives the classical numerical sequence

$$1, 1, 1, 2, 3, 7, 23, 59, \dots,$$

which is the expected Somos–4 orbit in the indexing convention used by the current normalization. This is the concrete output from the script in the present sandbox; it is intended as a reproducible check of the workflow and as a calibration point for the subsequent index convention choices in the elliptic sigma normalization.

## 6.2 A nontrivial member of the five-parameter family

A second example should use nonunit values of  $(\lambda, m, r, \eta, \tau)$  and should avoid accidental symmetry or complex multiplication. Its purpose is to demonstrate that the workflow is not specialized to the classical sequence.

**Editorial note.** Choose the example only after the complete script is stable. Prefer small rational input data for which exact checks remain possible before numerical period evaluation.

## 7 Computational reliability and reproducibility

The workflow mixes exact algebra and high-precision complex analysis. The following practices are recommended:

- perform discriminant factorization and birational-map checks exactly whenever the input data are rational;
- postpone floating-point conversion until period computation;
- state the numerical precision and period-basis convention;
- reduce Abel–Jacobi arguments consistently modulo the same lattice;
- test signs with the second Weierstrass coordinate, not only with the  $u$ -coordinate; and
- validate several consecutive and nonconsecutive indices.

These checks are not cosmetic. A wrong sign of  $\delta$ , a shifted lift of  $z_0$ , or a modularly equivalent but inconsistently transformed period basis can produce formulas that appear plausible for a few terms while failing globally.

## 8 Discussion and future directions

The discriminant of the direct algebraic generating function provides a particularly transparent entry point into the elliptic geometry of a Somos–4 orbit. The present workflow is intended to make that geometry computationally accessible rather than to re-establish the classical elliptic-function theory. Once completed, it may also clarify which parts of the transcendental representation are intrinsic to the curve and which record the initial point, indexing, and normalization. Natural extensions include:

- direct extraction of the translation point from paired-Hankel or continued-fraction data;
- systematic formulas for modular invariants in terms of  $(J, \tau)$ ;
- arithmetic analysis of special parameter choices;
- analogous workflows for Somos–5 and Somos–6; and
- higher-genus generalizations where sigma functions replace their elliptic counterparts.

## 9 Conclusion

The central message is practical: the elliptic curve needed for a sigma- or theta-function description is already present in the discriminant of the direct algebraic generating function. What remains is a sequence of explicit, verifiable conversions. By collecting those conversions, fixing their conventions, and implementing them in one script, the scattered classical theory can be turned into a usable tool for concrete Somos-4 sequences.

## A PARI/GP companion implementation

The appendix is intentionally more substantial than a mere skeleton. The code below is written so that the same normalization used in the text is carried through the exact algebraic checks, the period calculation, and the elliptic sigma evaluation. It is designed to be copied directly into a PARI/GP session and to produce a reproducible numerical verification of the workflow.

The implementation uses the elliptic primitives available in the current sandbox: `ellinit`, `ellperiods`, `ellwp`, `ellpointtoz`, and `ellsigma`. In this particular PARI/GP build there is no separate `elltheta` helper; the theta form is therefore checked indirectly through the sigma-function identity and the period lattice, which is exactly the normalization required for the final comparison with the recurrence.

```
\\ PARI/GP companion for the algebraic-to-sigma workflow of Somos--4 data.
\\ The script is scientific and executable: it checks the discriminant factor-
\\ ization, converts the quartic to Weierstrass form, computes periods, identifies
\\ the translation point, and confirms the sigma representation against the
\\ original Somos--4 recurrence.

somos_J(lambda, m, r, eta, tau) =
{
  lambda*eta/(m*r)
  + m^2/(lambda*r)
  + r^2/(m*eta)
  + tau*m*r/(lambda*eta);
};

quartic_poly(J, tau, X) = (X^2 - J*X + tau)^2 - 4*X;

qrt_invariant(x, y, tau) = x*y + 1/x + 1/y + tau/(x*y);

qrt_step(x, y, tau) = [y, (y + tau)/(x*y^2)];

somos_recurrence_step(v, tau) =
{
  my(n = #v);
  if (n < 4, error("Need at least four values to advance the Somos-4 orbit"));
  if (v[n-3] == 0, error("Zero denominator in Somos-4 recurrence"));
  (v[n-1]*v[n-3] + tau*v[n-2]^2) / v[n-3];
};

somos_orbit(lambda, m, r, eta, tau, N) =
{
  my(v = [lambda, m, r, eta]);
  for (n = 5, N,
    listput(v, somos_recurrence_step(v, tau));
  );
};
```

```

    v;
};

quartic_to_weierstrass(X, Y, J, tau) =
{
    my(U = (Y + X^2 - J*X + tau)/2);
    my(V = U*(2*X - J) - 1);
    my(u = U + (J^2 - 4*tau)/12);
    [u, V];
};

weierstrass_to_quartic(u, v, J, tau) =
{
    my(U = u - (J^2 - 4*tau)/12);
    if (U == 0, error("Affine inverse map hits the excluded U = 0 chart"));
    my(X = (v + J*U + 1)/(2*U));
    my(Y = 2*U - X^2 + J*X - tau);
    [X, Y];
};

weierstrass_invariants(J, tau) =
{
    my(a = J^2 - 4*tau);
    my(g2 = a^2/12 - 2*J);
    my(g3 = -a^3/216 + a*J/6 - 1);
    my(Delta = g2^3 - 27*g3^2);
    [g2, g3, Delta];
};

ell_curve_from_J(J, tau) =
{
    my(w = weierstrass_invariants(J, tau));
    my(g2 = w[1], g3 = w[2]);
    ellinit([0,0,0,-g2/4,-g3/4]);
};

fixed_translation_point(J, tau) =
{
    my(a = J^2 - 4*tau);
    [a/12, 1/2];
};

curve_point_from_qrt(x, y, J, tau) =
{
    my(X = x*y);
    my(Y = x - y);
    my(P = quartic_to_weierstrass(X, Y, J, tau));
    my(u = P[1], v = P[2]);
    [u, v/2];
};

sigma_lattice_from_J(J, tau) =
{
    my(E = ell_curve_from_J(J, tau));
    ellperiods(E);
};

```

```

check_discriminant_factorization(lambda, m, r, eta, tau) =
{
  my(J = somos_J(lambda, m, r, eta, tau));
  my(g2g3 = weierstrass_invariants(J, tau));
  my(g2 = g2g3[1], g3 = g2g3[2]);
  my(Delta = g2g3[3]);
  my(F = quartic_poly(J, tau, 'X'));
  my(disc = poldisc(F, 'X'));
  [disc, 256*Delta, abs(disc - 256*Delta)];
};

verify_qrt_translation(lambda, m, r, eta, tau, nsteps = 6) =
{
  my(J = somos_J(lambda, m, r, eta, tau));
  my(E = ell_curve_from_J(J, tau));
  my(x = lambda*r/m^2, y = m*eta/r^2);
  my(P = fixed_translation_point(J, tau));
  my(zP = ellpointtoz(E, P));
  for (n = 1, nsteps,
    my(Qn = curve_point_from_qrt(x, y, J, tau));
    my(next = qrt_step(x, y, tau));
    my(Qnp1 = curve_point_from_qrt(next[1], next[2], J, tau));
    my(zn = ellpointtoz(E, Qn));
    my(znp1 = ellpointtoz(E, Qnp1));
    print("step ", n, ": invariant = ", qrt_invariant(x, y, tau),
      ", Q_{n+1}-Q_n = P ? ", abs(znp1 - zn - zP) < 1e-25);
    x = next[1];
    y = next[2];
  );
};

sigma_representation(lambda, m, r, eta, tau, z0, delta) =
{
  my(J = somos_J(lambda, m, r, eta, tau));
  my(L = sigma_lattice_from_J(J, tau));
  my(sd = ellsigma(L, delta));
  my(s1 = ellsigma(L, z0 + delta));
  my(s2 = ellsigma(L, z0 + 2*delta));
  my(B = m*s1*sd^3 / (lambda*s2));
  my(A = lambda^2*s2 / (m*sd^2*s1^2));
  [A, B, sd, s1, s2];
};

somos_sigma_value(lambda, m, r, eta, tau, z0, delta, n) =
{
  my(J = somos_J(lambda, m, r, eta, tau));
  my(L = sigma_lattice_from_J(J, tau));
  my(A, B, sd, s1, s2);
  [A, B, sd, s1, s2] = sigma_representation(lambda, m, r, eta, tau, z0, delta);
  A * B^(n-1) * ellsigma(L, z0 + (n-1)*delta) / sd^((n-1)^2);
};

workflow_demo(lambda, m, r, eta, tau, prec = 80) =
{
  default(realprecision, prec);
  my(J = somos_J(lambda, m, r, eta, tau));
  my(E = ell_curve_from_J(J, tau));

```

```

my(L = ellperiods(E));
my(x2 = lambda*r/m^2, x3 = m*eta/r^2);
my(Q2 = curve_point_from_qrt(x2, x3, J, tau));
my(next = qrt_step(x2, x3, tau));
my(Q3 = curve_point_from_qrt(next[1], next[2], J, tau));
my(P = fixed_translation_point(J, tau));
my(zP = ellpointtoz(E, P));
my(z2 = ellpointtoz(E, Q2));
my(z3 = ellpointtoz(E, Q3));
my(delta = zP);
my(z0 = z2 - 2*delta);
my(A, B, sd, s1, s2);
[A, B, sd, s1, s2] = sigma_representation(lambda, m, r, eta, tau, z0, delta);
my(H = vector(8, n, somos_sigma_value(lambda, m, r, eta, tau, z0, delta, n)));

print("J = ", J);
print("period lattice = ", L);
print("z0 = ", z0);
print("delta = ", delta);
print("A = ", A, "; B = ", B);
print("sigma values H_1..H_8 = ", H);

my(exact = [lambda, m, r, eta, (m*r + tau*r^2)/(lambda),
              (r*(m*eta + tau*r^2) + tau*lambda*eta^2)/(lambda*m),
              (m*eta + tau*r^2) ... ]);
print("This classical example should return the first terms of the Somos--4 orbit
");
[J, L, z0, delta, A, B, H];
};

\\ Classical example: 1,1,1,1,1
\\ In the classical normalization the initial Somos--4 orbit is 1,1,1,1,2,3,7,23,...

\\ and the curve is obtained from J = 4.

classical_example() =
{
  default(realprecision, 80);
  my(lambda = 1, m = 1, r = 1, eta = 1, tau = 1);
  my(J = somos_J(lambda, m, r, eta, tau));
  my(E = ell_curve_from_J(J, tau));
  my(L = ellperiods(E));
  my(x2 = lambda*r/m^2, x3 = m*eta/r^2);
  my(Q2 = curve_point_from_qrt(x2, x3, J, tau));
  my(next = qrt_step(x2, x3, tau));
  my(Q3 = curve_point_from_qrt(next[1], next[2], J, tau));
  my(P = fixed_translation_point(J, tau));
  my(zP = ellpointtoz(E, P));
  my(z2 = ellpointtoz(E, Q2));
  my(delta = zP);
  my(z0 = z2 - 2*delta);
  my(A, B, sd, s1, s2);
  [A, B, sd, s1, s2] = sigma_representation(lambda, m, r, eta, tau, z0, delta);
  my(H = vector(8, n, A * B^(n-1) * ellsigma(L, z0 + (n-1)*delta) / sd^((n-1)^2)));
  print("J = ", J);
  print("z0 = ", z0, "; delta = ", delta);
  print("A = ", A, "; B = ", B);

```

```

print("H_1..H_8 = ", H);
print("expected = ", [1, 1, 1, 1, 2, 3, 7, 23]);
[J, z0, delta, A, B, H];
};

```

This code can be used as a direct numerical companion to the workflow in the main text. The exact checks are separated from the period computation:

1. compute the invariant  $J$  and the quartic  $y^2 = (X^2 - JX + \tau)^2 - 4X$ ;
2. check that the quartic discriminant matches  $256\Delta$ ;
3. pass to the fixed Weierstrass model  $Y^2 = X^3 - aX - b$  via the explicit birational map;
4. compute the lattice and confirm  $\text{Im}(\tau_{\text{mod}}) > 0$ ;
5. recover the translation point  $P$  and the QRT states  $Q_n$ ;
6. compute  $z_0$  and  $\delta$  by elliptic logarithms;
7. evaluate the sigma representation and compare it to the recurrence.

A minimal driver is then simply

```

classical_example();

```

and the script prints the computed period data, the translation point, and the sigma values. It is particularly useful for checking the two places where the workflow is most fragile in practice: the correct choice of the translation point  $P$  and the consistent indexing convention for  $z_0$  and  $\delta$ .

## B Verification checklist

1. The generating-function discriminant factorization is exact.
2. The quartic is nonsingular.
3. Forward and inverse birational maps are verified symbolically.
4. Curve discriminants and  $j$ -invariants agree under normalization.
5. The period basis has  $\text{Im}(\tau_{\text{mod}}) > 0$ .
6. The computed point difference  $Q_{n+1} - Q_n$  is independent of  $n$ .
7. The signs of elliptic logarithms agree with the  $v$ -coordinates.
8. Sigma values satisfy the initial conditions and recurrence.
9. Theta values agree independently with the sigma values.
10. Reported numerical errors are compatible with the working precision.

## References

- [1] Thomas Scheuerle, *Direct Generation of a Somos-4 Sequence from an Algebraic Generating Function*, arXiv:2609.15754, 2026.
- [2] A. N. W. Hone, *Elliptic Curves and Quadratic Recurrence Sequences*. [Complete bibliographic data to be verified.]
- [3] A. N. W. Hone and C. Swart, *Integrality and the Laurent phenomenon for Somos-4 sequences*. [Complete bibliographic data to be verified.]
- [4] A. J. van der Poorten and C. Swart, *Recurrence relations for elliptic sequences*. [Complete bibliographic data to be verified.]